

Dr Benya Wang

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CAREER PROFILE

AI Architect with a PhD and 8 years of experience designing, governing, and scaling enterprise-grade data science and AI platforms across mining and resources. Sets enterprise AI standards, reference architectures, and roadmaps, then guides delivery teams to build against them through platform patterns, solution design review, and mentoring across business units. Deep Azure specialisation (Azure AI Foundry, Azure OpenAI, Azure ML) alongside Databricks, LLMops and MLOps tooling, with a track record of taking Generative AI, RAG knowledge bases, and agentic systems from proof of concept to secure production. Designs AI and security governance frameworks with Cyber and Enterprise Architecture, and translates AI capability into strategic priorities for business leaders.

CORE COMPETENCIES

AI Architecture & Governance

AI governance framework design, secure-by-design patterns, data architecture and governance

Generative AI & LLMs

Azure OpenAI, Azure AI Foundry, Azure AI Search, Microsoft Copilot Studio, LangChain, LightRAG, GraphRAG, Claude Code, RAG pipeline design, vector stores, multi-agent orchestration, agentic platforms

Azure & Data Platforms

Azure (ACR, IoT Edge/Hub, VNet, Azure ML), Databricks (data warehouse, workflow orchestration), AWS

LLMOps, MLOps & DevOps

MLflow, Azure DevOps / Pipelines, Terraform, Bicep, Docker, AKS, ACI (containerised microservices), model endpoint deployment, CI/CD pipeline design

ML & Data Engineering

Python, PySpark, Pandas, SQL, XGBoost, LightGBM, predictive analytics

PROFESSIONAL EXPERIENCE

Specialist Machine Learning Engineer

Apr 2023 – Present

South32 · Perth, WA

- Led technical governance for the ML platform including authoring standards, engineering guides, and platform patterns; reviewing architectures; and mentoring data scientists across business units to drive consistent, production-quality delivery.
- Defined the technical strategy and built a unified AI platform (Azure AI Search, Azure OpenAI, Cosmos DB) with self-service patterns and reference implementations that reduced time-to-production for new LLM applications by ~60%, enabling multiple product teams to ship independently.
- Built a safety chatbot and GenAI knowledge base over safety and geoscience corpora (Azure AI Foundry, Azure AI Search, LightRAG, Azure Web App), giving operational teams conversational access to guidance previously locked in unstructured documents.
- Right-sized the container runtime for model inference endpoints — ACI for lightweight, single-service workloads and AKS where orchestration and scaling were required — packaging models as Docker microservices published through ACR.
- Pioneered an air-gapped edge ML deployment pattern (Azure IoT Edge, Databricks Apps) enabling model inference on isolated industrial sites with zero internet connectivity; adopted as the standard across all remote operations.
- Partnered with Cyber and Enterprise Architecture teams to design a Microsoft Copilot Studio governance framework and a GitHub / GitHub Copilot governance framework, establishing enterprise-wide standards for AI-assisted development, access controls, and compliant deployment; supported the organisational migration from Azure DevOps to GitHub at scale.

- Served as Acting Data Science Lead May–Jun 2026, maintaining delivery continuity across active AI platform workstreams; co-designed the platform production support model with the Operational Digital Technology team, defining ownership, escalation paths, and SLA structures.

Specialist Data Scientist

Jan 2022 – Apr 2023

South32 · Perth, WA

- Architected and led delivery of a centralised Data Science platform (Azure Databricks, MLflow, Azure Pipelines) that standardised MLOps workflows across 6+ business units, cutting the average model deployment cycle from ~3 weeks to under 5 days and eliminating ad-hoc deployment risk.
- Delivered advanced process control solutions for operations in Australia, Africa, and South America, working with local technology teams at each site to improve recovery and energy efficiency.
- Built reusable PySpark feature pipelines into a Databricks Feature Store over large-scale industrial sensor data, replacing ad-hoc notebook preparation with versioned, model-ready features.
- Partnered with data governance teams to implement stewardship and metadata management aligned with enterprise compliance policies.
- Designed and deployed a Vision AI safety system (YOLO, Supervision) for automated stop-sign monitoring, directly improving operational safety compliance.
- Implemented production multi-agent LLM pipelines (GraphRAG, AutoGen) to extract structured insights from HR records, safety incident reports, and geoscience exploration corpora.

Data Scientist

Jan 2021 – Jan 2022

Rio Tinto · Perth, WA

- Engineered serverless data pipelines (AWS Lambda, Glue, Athena) for real-time production analytics supporting operational decision-making.
- Led ML-driven optimisation of ore blending strategies, improving resource utilisation and reducing annual operational costs.
- Won an internal Hackathon for diamond image classification using TensorFlow and transfer learning.

Graduate Data Scientist

Feb 2019 – Jan 2021

Rio Tinto · Perth, WA

- Developed predictive maintenance models (LightGBM, Random Forest) for a 1,700 km rail network, reducing unplanned downtime through early fault detection.
- Containerised models in Docker and deployed via AWS SageMaker, establishing repeatable deployment and rollback processes for the team.

EDUCATION

PhD in Environmental Engineering

Feb 2015 – Feb 2019

University of Western Australia

Thesis: Application of machine learning algorithms in urban catchments. Five peer-reviewed publications in *Geoscientific Model Development*, *Water Resources Research*, and other journals.

Bachelor of Science

Sep 2010 – Jun 2014

Zhejiang University, China

CERTIFICATIONS

Databricks Certified

Machine Learning Professional · Machine Learning Associate · Data Engineer Associate

Microsoft Certified

DevOps Engineer Expert · Azure Data Scientist Associate · Azure Data Engineer Associate · Azure Developer Associate

AWS Certified

Machine Learning – Specialty · Solutions Architect – Associate

Other

HashiCorp Certified: Terraform Associate · Anthropic: Claude Code in Action